

Anti-Lock Brake System (ABS) and Stability Control



AdvanceTrac® With RSC®

The AdvanceTrac® system is made up of the following systems:

- Anti-lock braking
- Traction control
- ESC

The RSC® system is an optional feature that works in conjunction with AdvanceTrac® to aid in maintaining vehicle stability. AdvanceTrac® with RSC® system consists of the following components:

- ABS module — attached to the Hydraulic Control Unit (HCU) , but can be serviced separately from the HCU .
- Front wheel speed sensors — one sensor is located in each front wheel knuckle. The sensors are serviced separately from the knuckle.
- Front wheel speed sensor encoders — one encoder strip is located in each wheel bearing and wheel hub assembly and is serviced with the wheel bearing.
- HCU — contains several valves (inlet, outlet, isolation and dump) and a hydraulic pump motor. It is mounted behind the radiator, below the air cleaner assembly. The HCU is replaced as a unit, including the ABS module.
- Rear wheel speed sensors — one sensor is located in each end of the rear axle housing. The sensors are serviced separately.
- Rear wheel speed sensor tone rings — one tone ring is located on each axle shaft.
- Stability control sensors — located inside the Restraints Control Module (RCM) and consist of the following sensors: yaw rate, roll rate, lateral and longitudinal accelerometers. The sensors are integral to the RCM and cannot be serviced separately.
- Steering wheel rotation sensor (6.2L) — located on the end of the steering column near the intermediate shaft. The sensor is serviced separately from the steering column but the sensor ring is replaced with the column. For all other applications, the steering wheel rotation sensor is part of the EPAS .
- Stability/traction control switch — located in the instrument panel finish panel and is serviced separately from the finish panel.
- Hill Descent Control™ switch — located in the auxiliary switch panel and is serviced separately.
- Brake vacuum pump (3.5L) — a mechanical pump located on the back of the RH cylinder head and driven by the RH exhaust cam shaft.

Electronic Brake Distribution (EBD)

The ABS module and HCU incorporate a strategy called EBD . The EBD strategy uses the HCU as an electronic proportioning valve to create a balanced braking condition between the front and rear wheels and minimizes the chance of rear wheel lockup during hard braking.

A slight bump sensation may be felt in the brake pedal when EBD is active.

Anti-Lock Braking

The conventional ABS aids in the prevention of wheel lock-up during braking events allowing the driver to maintain steering control and stop in the shortest distance possible under most conditions. This is accomplished by the ABS module and the HCU modulating brake fluid pressure to the affected brake caliper(s).

When the ABS activates, both of the following can occur:

- A rumble or grinding sound may be heard

- If the driver's foot is on the brake pedal, a vibration may be felt in the brake pedal

Supplemental Braking Assist

In addition to preventing wheel lock up during braking events, the ABS module also provides supplemental hydraulic brake assist through the use of the hydraulic pump motor and the **HCU**. This is done in the event of a severe vacuum loss at the brake booster.

AdvanceTrac® System

Traction Control System

The traction control system aids in the prevention of excessive wheel spin allowing the vehicle to maintain traction during acceleration. This is accomplished by the ABS module and the **HCU** modulating brake fluid pressure to the affected brake caliper(s) and by the PCM modulating engine torque.

Electronic Stability Control (ESC) System

The **ESC** system constantly monitors the vehicle's direction of travel relative to the driver's intended course, except when the vehicle is traveling in reverse, and aids in keeping the vehicle traveling along that course. This is accomplished by the ABS module and the **HCU** modulating brake fluid pressure to the affected brake caliper(s) and by the PCM modulating engine torque.

When the AdvanceTrac® system activates, any of the following can occur:

- A rumble or grinding sound may be heard, much like ABS
- If the driver's foot is on the brake pedal, a vibration may be felt in the brake pedal, much like ABS
- A small deceleration or a reduction in the acceleration of the vehicle
- The stability/traction control indicator (sliding car icon) flashes

Roll Stability Control (RSC®) System

The **RSC®** system constantly monitors vehicle roll motion, except when the vehicle is traveling in reverse, and aids in keeping the vehicle stable during sudden or abrupt maneuvers. This is accomplished by the ABS module and the **HCU** modulating brake fluid pressure to the affected brake caliper(s) and by the PCM modulating engine torque. The combination of reduced engine torque and modulated brake fluid pressure aids in reducing the cornering forces and, therefore, the total roll momentum acting on the vehicle.

When the **RSC®** system activates, the vehicle produces the same noticeable symptoms as it does when the AdvanceTrac® system activates.

Trailer Sway Control

Trailer sway control is a unique function in the AdvanceTrac® with **RSC®** system that recognizes and aids in reducing trailer sway. The trailer sway control function continually monitors the steering wheel rotation and yaw rate sensors to identify specific, increasing yaw oscillations that occur during a trailer sway event. If the trailer sway event exceeds programmed parameters, the ABS module and the **HCU** modulate brake fluid pressure to specific brake caliper(s) and the PCM modulates engine torque until the trailer sway is reduced below the programmed parameters.

For vehicles equipped with an optional **TBC** module, the trailer sway control system utilizes the module to apply the trailer brakes to quickly reduce trailer sway.

When the trailer sway control function activates, the vehicle produces the same noticeable symptoms as it does when the AdvanceTrac® system activates. In addition, the message center displays TRAILER SWAY, REDUCE SPEED and, if the vehicle is equipped with a **TBC** module, the module shows output to the trailer brakes.

The trailer sway control function only activates when the vehicle speed exceeds 64 km/h (40 mph). The system can be disabled during any key cycle. Refer to the Owner's Literature for information on disabling the system. The system is re-enabled when the ignition is turned to the OFF position and back to the RUN position.

MyKey® Interaction

Through the MyKey® feature, the traction control function of the AdvanceTrac® system can be configured to be always on or to allow the driver to select the traction control function on or off.

When the traction control function is configured to be always on and a MyKey® restricted key is in use, the **IPC** will ignore any requests made by the driver to disable the traction control function and will not send any traction control disable messages to the ABS module. Refer to the Owner's Literature for additional information on the MyKey® feature and settings.

Hill Start Assist

The hill start assist system is designed to assist the driver during hill-starts. Using the ABS, the hill start assist system holds the vehicle on an incline for a short time, allowing the driver to release the brake pedal and press the accelerator pedal without needing to use the parking brake.

Hill Descent Control™

Hill Descent Control™ maintains vehicle speed on a downhill grade, operating at speeds between 3 km/h (2 mph) and 32 km/h (20 mph). If the vehicle is on flat or uphill terrain, the system goes into standby mode, remaining activated up to 64 km/h (40 mph). At speeds greater than 64 km/h (40 mph), the Hill Descent Control™ system is deactivated. When Hill Descent Control™ is active, a rumble or grinding sound may be heard and/or felt.

Traction Control Switch

The stability/traction control switch can be used by the driver to disable and enable the traction control, **ESC** and **RSC®** functions. Refer to the Owner's Literature for instructions on disabling and enabling the traction control, **ESC** and **RSC®** functions.

The conventional ABS cannot be disabled using the stability/traction control switch.

Stability/Traction Control Indicators

The **IPC** message center is a reconfigurable telltale which is capable of displaying a variety of different messages and indicators such as the stability/traction control indicator, stability/traction control disabled indicator, the **4WD** mode/range or door open messages. For information about the **IPC**, refer to Section 413-01.

There are 2 separate stability/traction control indicators, the sliding car icon and the sliding car OFF icon. Both icons are located in the **IPC** message center. The stability/traction control indicator is used to alert the driver that a stability/traction event is taking place (flashes twice per second) and to alert the driver of potential concerns in the AdvanceTrac® system (illuminates continuously). The sliding car OFF icon is used to alert the driver that the traction control portion of the AdvanceTrac® system has been disabled by the driver. This indicator only illuminates continuously and does not flash.

One or both of the stability/traction control indicators may illuminate as a result of momentary sensor disturbances due to environmental or driving conditions (including severe vehicle maneuvers or extreme off road usage). Once Illuminated, the indicator remains illuminated until the environmental or driving condition is no longer present and the ignition is cycled from ON to OFF and then back to ON again. If there are no other customer concerns, symptoms, indicators or **DTCs**, the stability/traction control indicator may have been illuminated due to these environmental or driving conditions.